Omkar Tanaji Patil

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LinkedIn | GitHub | LeetCode | CodeChef | Kaggle

Education

Pimpri Chinchwad College of Engineering and Research — Pune, India
Bachelor of Engineering in Computer Science; Aggregate GPA: 9.84 — *Current*

Kulbhushan Junior College — Chh. Sambhajinagar
Higher Secondary Certificate (HSC) — *Completed*

Ramchandra Naik School — Chh. Sambhajinagar
Secondary School Certificate (SSC) - Score: 97% — *Completed*

Experience

Google Developer Group (GDG) Students Club — Pune
DSA Facilitator — *Aug 2025 – Present*

- Conducted Data Structures and Algorithms workshops for 100+ students.
- Worked with the core team to organize technical events and activities.

Projects

Lungs Disease Diagnosis | *Computer Vision, Deep Learning*

- Developed a deep learning pipeline to classify chest X-ray images into four categories: Healthy, COVID-19, Viral Pneumonia, and Bacterial Pneumonia.
- Tech Used: Python, TensorFlow, Keras, CNNs, OpenCV, Matplotlib | **95% Classification Accuracy**

Student Performance Predictor | *Data Analysis, Regression Modeling*

- Analyzed academic datasets and engineered features to build robust regression models predicting student performance metrics.
- Tech Used: Python, Pandas, CatBoost, XGBoost, Docker, AWS | **88% R2 Score**

SwiftAnim - GenAI Animation Tool | *Data Visualization, GenAI*

- Engineered an automated data-to-video pipeline translating textual prompts and data into programmatic visual animations using Manim and LLMs.
- Tech Used: Python, ManimGL, OpenAI API (LLMs), Streamlit | **100% Automated Pipeline**

CineMatch - Movie Recommendation Engine | *Exploratory Data Analysis (EDA)*

- Processed and cleaned a 5,000+ record dataset to develop a content-based recommendation engine utilizing vector and cosine similarity metrics.
- Tech Used: Python, Pandas, Scikit-learn, Cosine Similarity | **Optimized for 100ms Query Latency**

Technical Skills

Languages: Python, SQL, C++, HTML, CSS
Libraries/Tools: Pandas, Polars, Scikit-learn, TensorFlow, PyTorch, NumPy, Matplotlib, Seaborn, Plotly, Flask
Databases/Cloud: MySQL, MongoDB, AWS (Elastic Beanstalk), Docker
Core Competencies: Data Analysis, Deep Learning, Statistical Modeling, Machine Learning, Exploratory Data Analysis (EDA), Data Structures & Algorithms

Achievements & Certifications

Competitive Programming: LeetCode Rating **1650+** (**500+ Solved**), CodeChef Rating **1440+**, **Kaggle** (5 Contests)
Hackathons: **Semi-Finalist** GenAI Exchange (**Google**), **3rd Prize** Craftverse National Hackathon, **SIH** (Internals)
Achievements: Won **1st Prize** CodeCraft DSA Contest, Achieved **10.0 SGPA** (2nd Year)
Certifications: **Data Science with GenAI** (PWSkills), **Machine Learning A-Z** (Udemy), Python (HackerRank), MongoDB Basics (Mongo Academy)